BABU BANARASI DAS UNIVERSITY

Session 2026-27



PREDICTIVE ANALYTICS

(BCADSN15301)

Submitted By: - Kirti Singh

Roll No.: - 1240258230

Batch: - BCADS34

Submitted To: - Mr. Deepak

PREDICTIVE ANALYTICS PROJECT – 2

Project Objective

The project imports the customer data, checks the dataset, selects the test-mailing records, sets the required field roles, and builds a CHAID model to predict customer response. The trained model is then used to score the remaining customers and prepare the final contact file.

The reference exercise describes a dataset containing 12 fields and 30,000 records. The modelling workflow uses gender and the RFM fields as inputs and response as the target. The screenshots below document the corresponding operations performed in your own stream.

Software Used: IBM SPSS Modeler

Input data:

- ACME customer and rfm data.sav

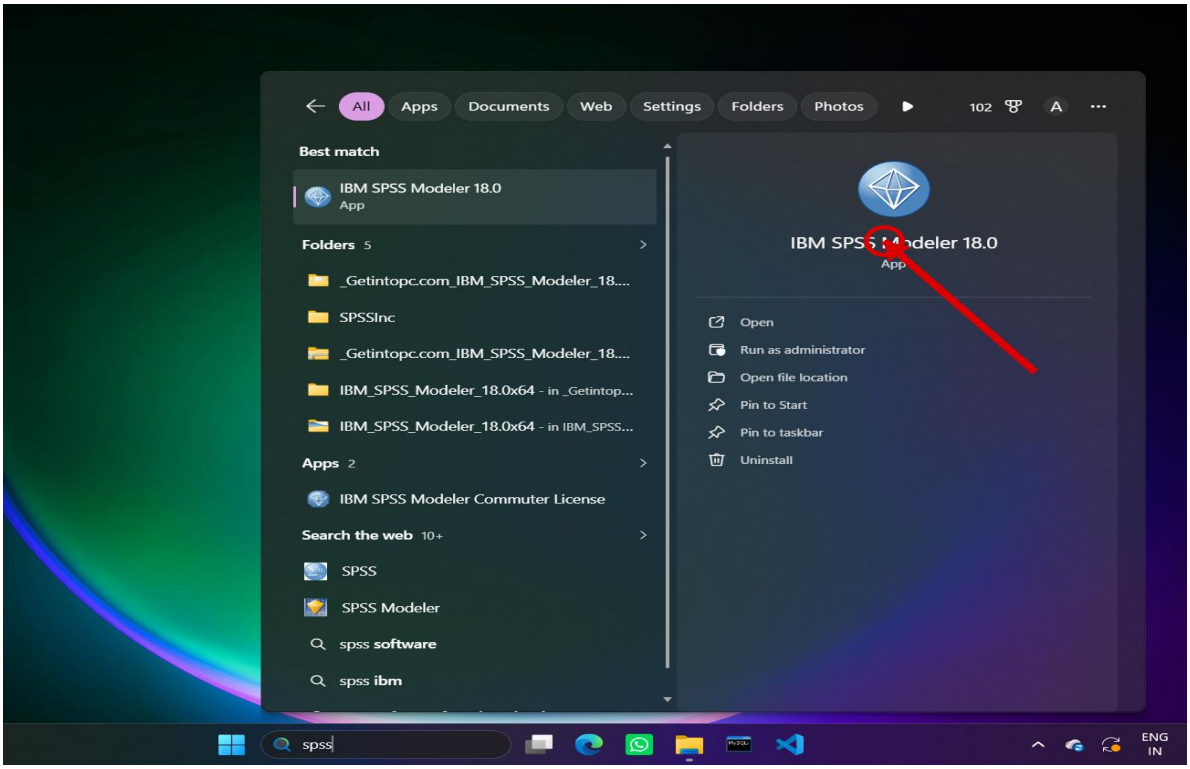
Working folder: C:\Users\kirti\OneDrive\Desktop\Vishal_IBM

Task 1 – Launch IBM SPSS Modeler and set the working directory

What to do: Start IBM SPSS Modeler, create the working stream, and set the project folder so that the input files are available from the correct location.

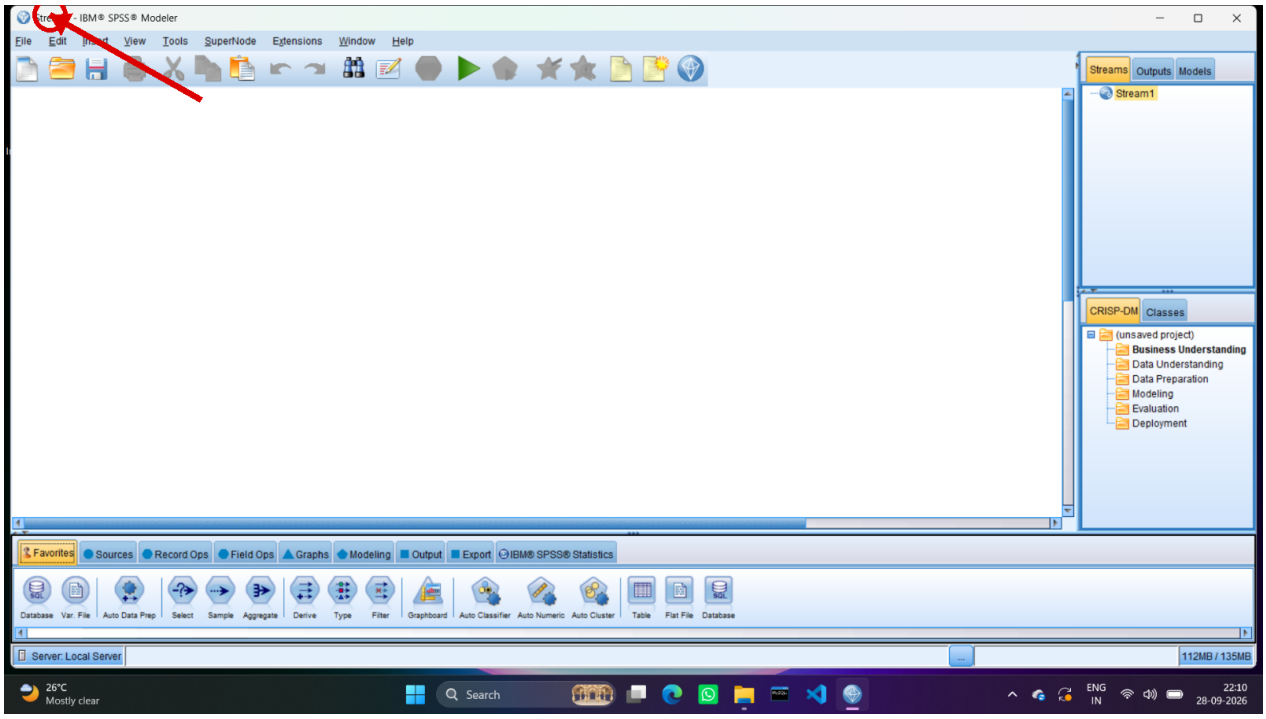
Step 1 – Launch IBM SPSS Modeler

Open the Windows search panel and select IBM SPSS Modeler 18.0. This starts the application and opens the Modeler workspace for creating the predictive analytics stream.



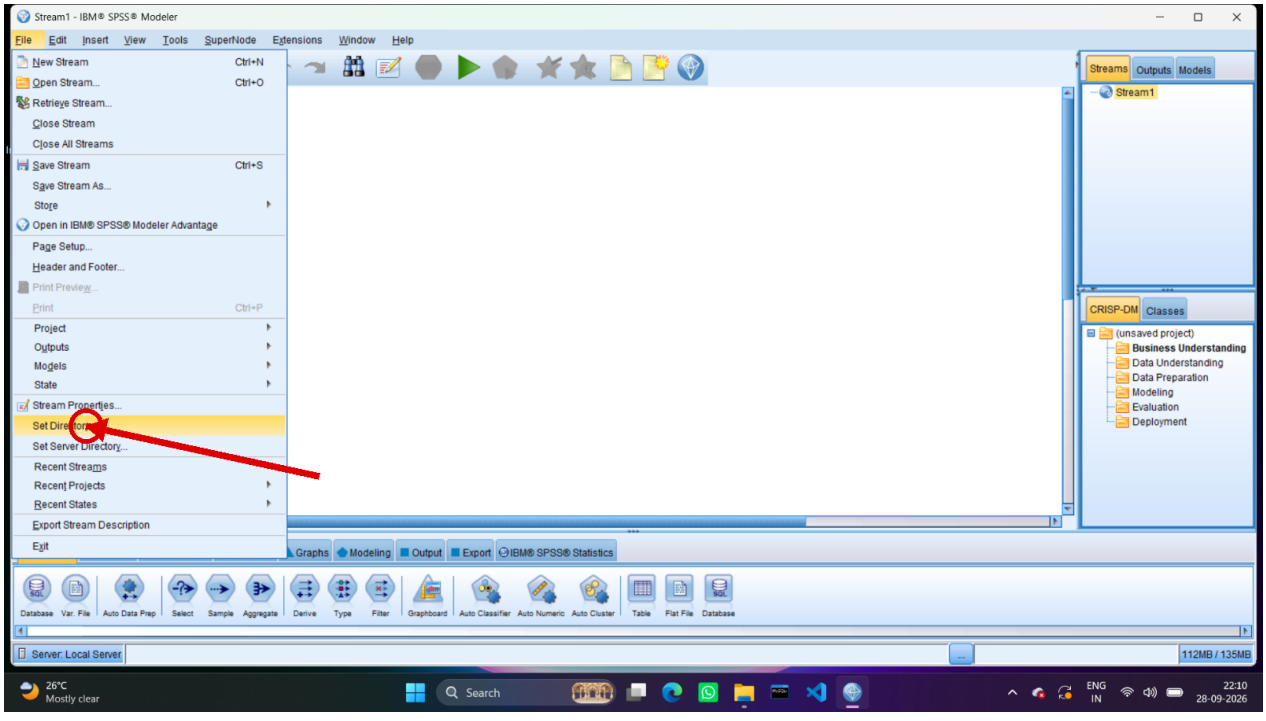
Step 2 – Open the Modeler stream workspace

After the application starts, the main Stream1 workspace is displayed. This blank canvas is used to place source, processing, modelling, and output nodes.



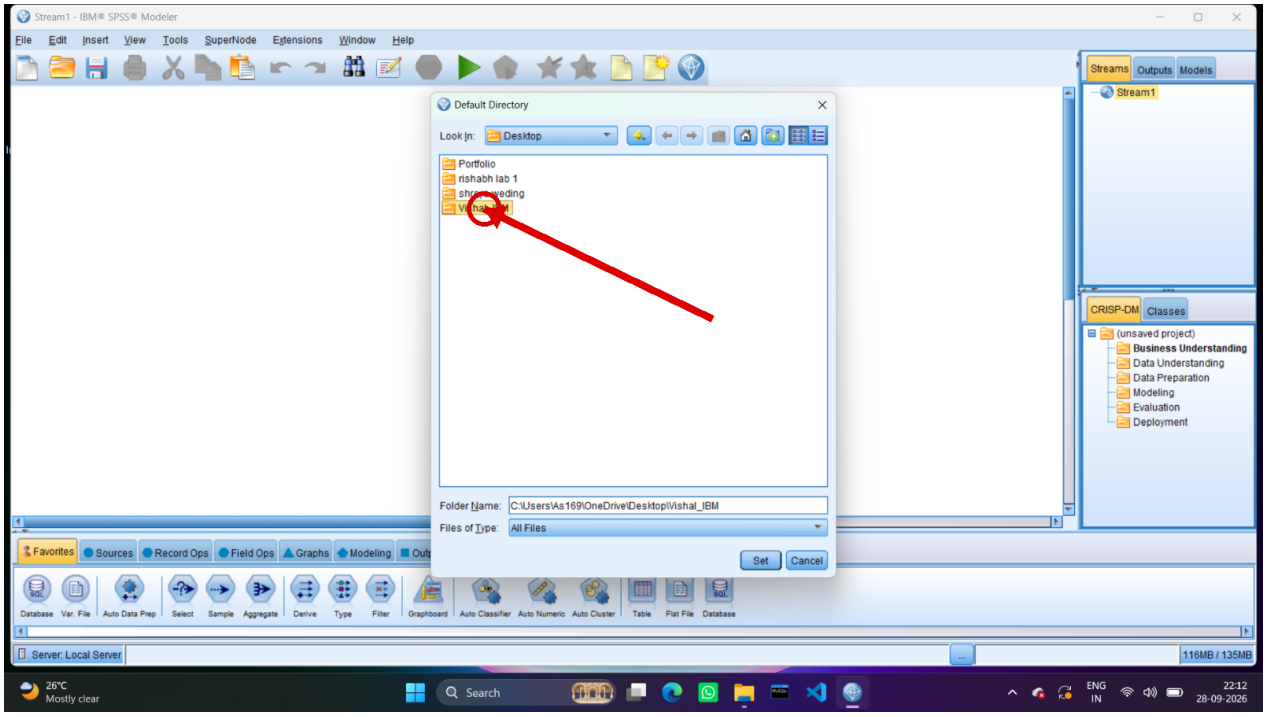
Step 3 – Open Set Directory

Open the File menu and choose Set Directory. This option is used to define the folder that will act as the working location for the project files.



Step 4 – Select the project folder

In the Default Directory window, select the kirti_IBM folder from the Desktop location. Confirm the folder so Modeler uses it as the working directory.

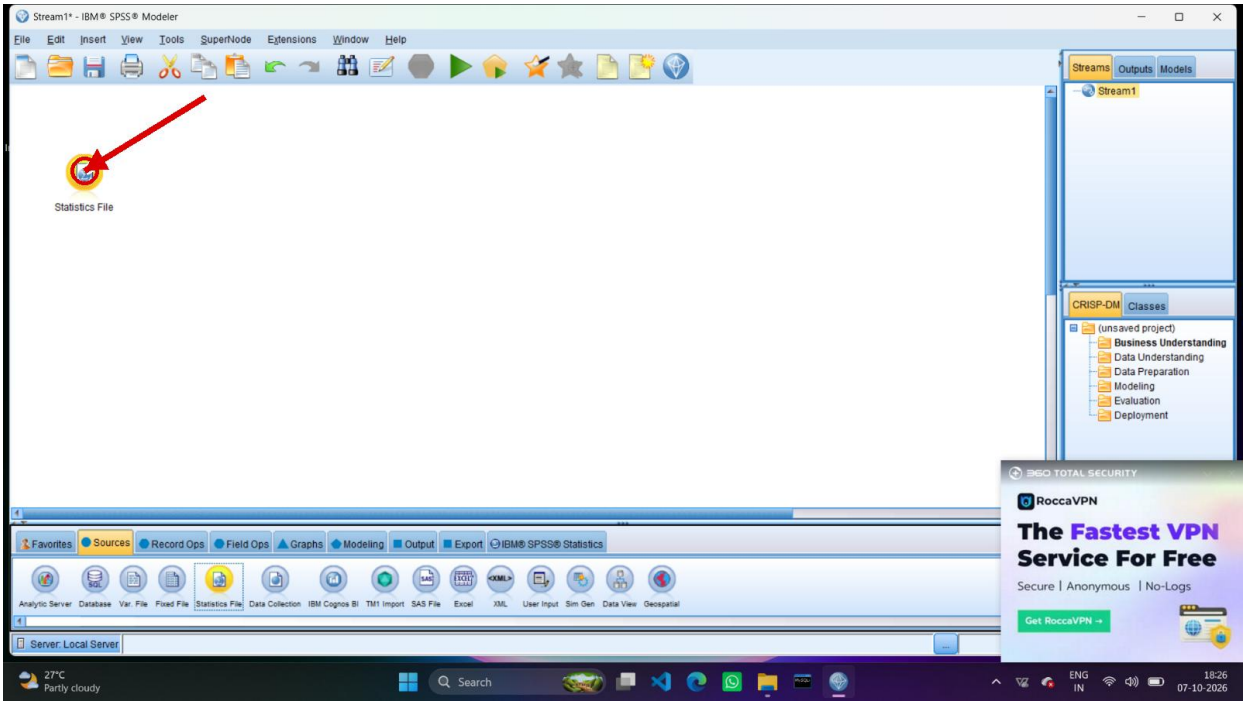


Task 2 — Import and examine the data

This stage brings the customer dataset into the stream and verifies that the imported records and fields are available for later modelling.

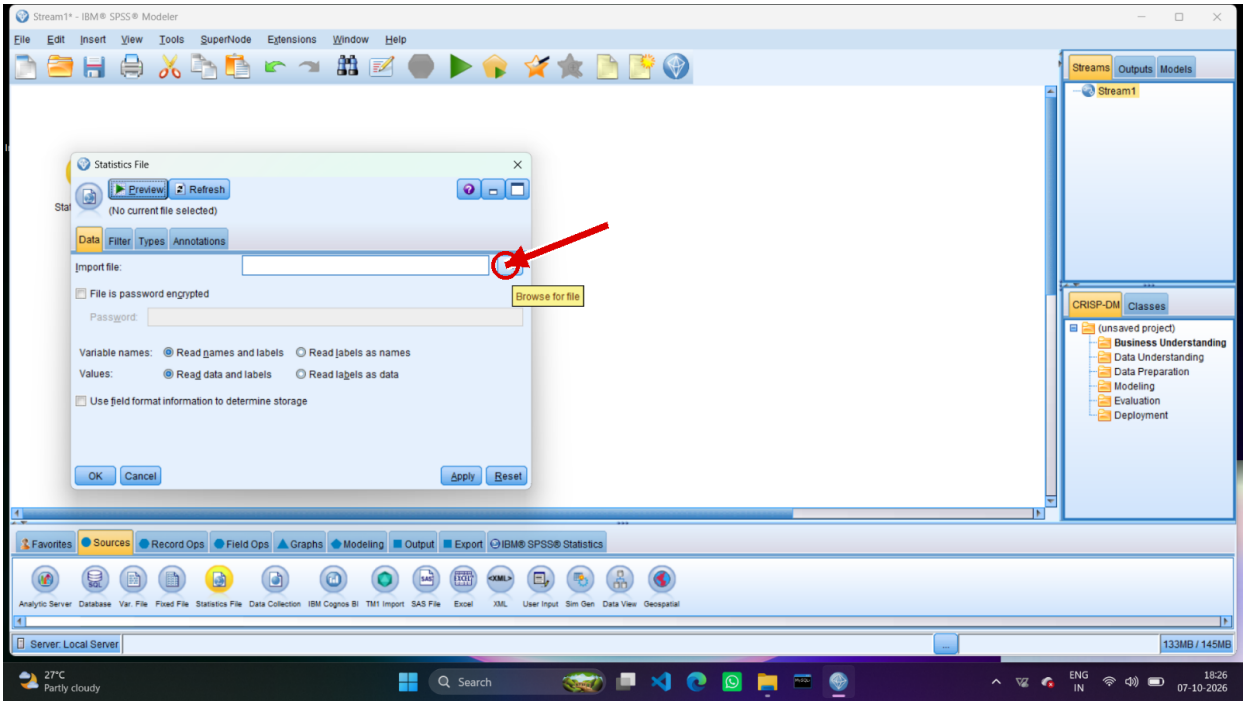
Step 5. Add the Statistics File source

Open IBM SPSS Modeler and use the Statistics File source from the Sources palette. This creates the starting point for importing the project dataset.



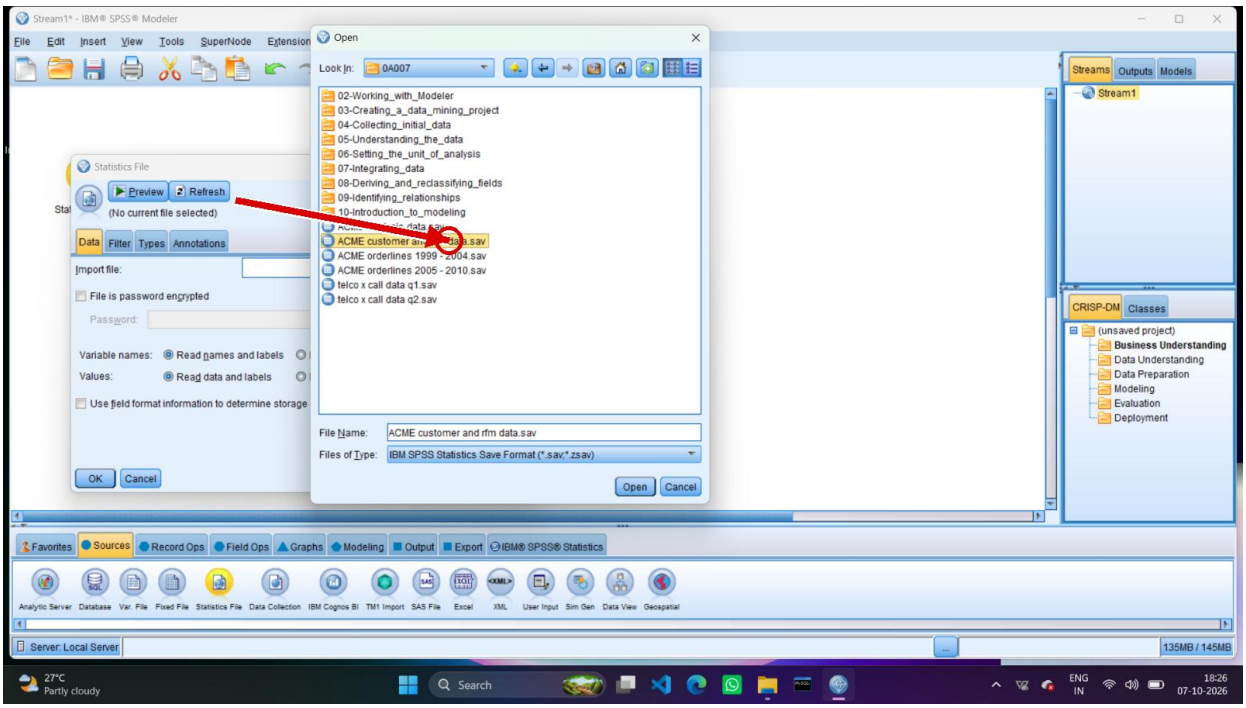
Step 6. Open the file browser

In the Statistics File settings, use Browse to locate the SPSS Statistics data file. Keep the normal/default import settings unless your project requires a change.



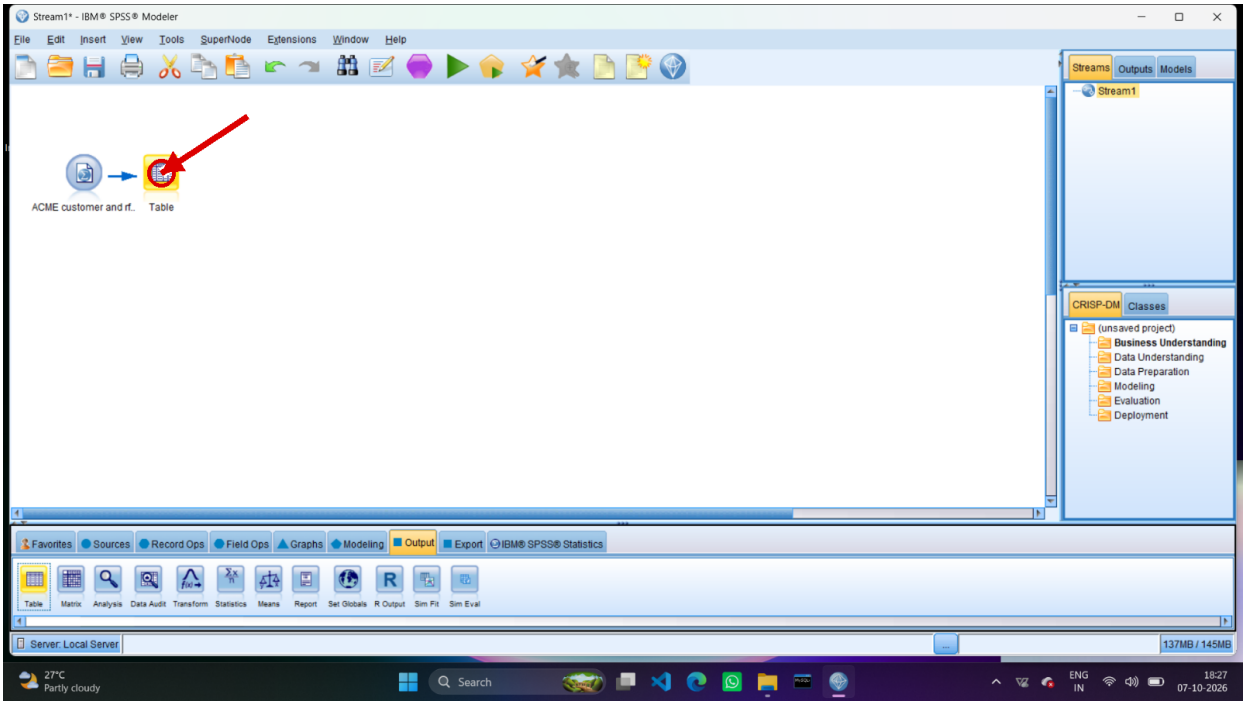
Step 7. Select the project data file

Select the required customer data .sav file in the folder and confirm the selection. The chosen file is then returned to the Statistics File node.



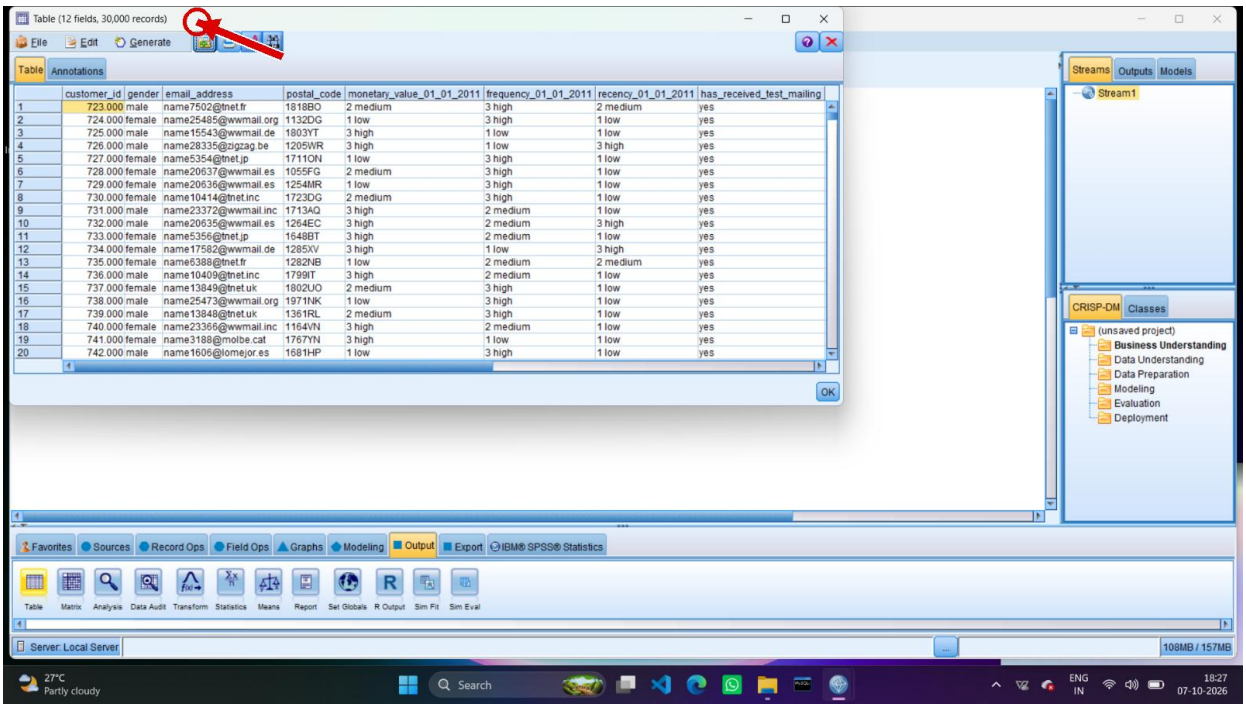
Step 8. Add a Table node

Connect a Table output to the imported data so the records can be inspected before modelling. The screenshot shows the source connected to the Table node.



Step 9. Examine the dataset

Run the Table node and inspect the returned data. The table window shows the dataset structure, including the 12-field and 30,000-record result required by the reference workflow.

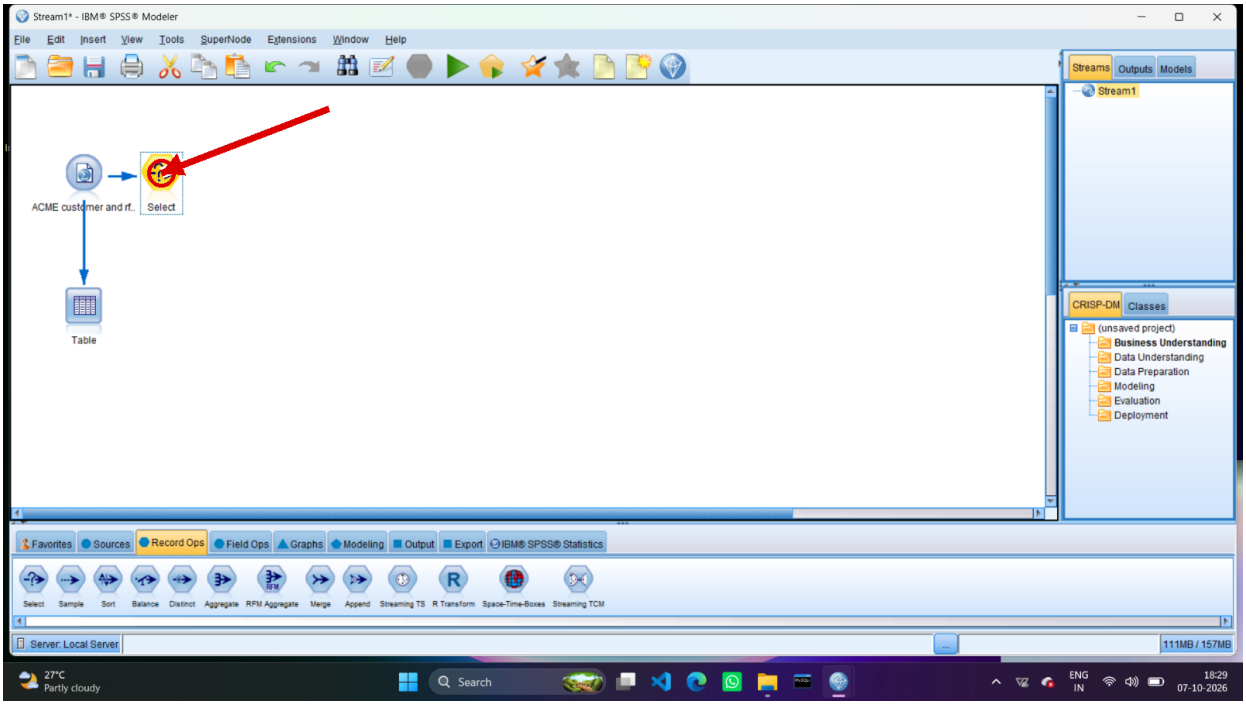


Task 3 — Select the data for modelling

Only the records required to train the response model should continue into the modelling branch. The Select and Table nodes provide both the filter and the verification.

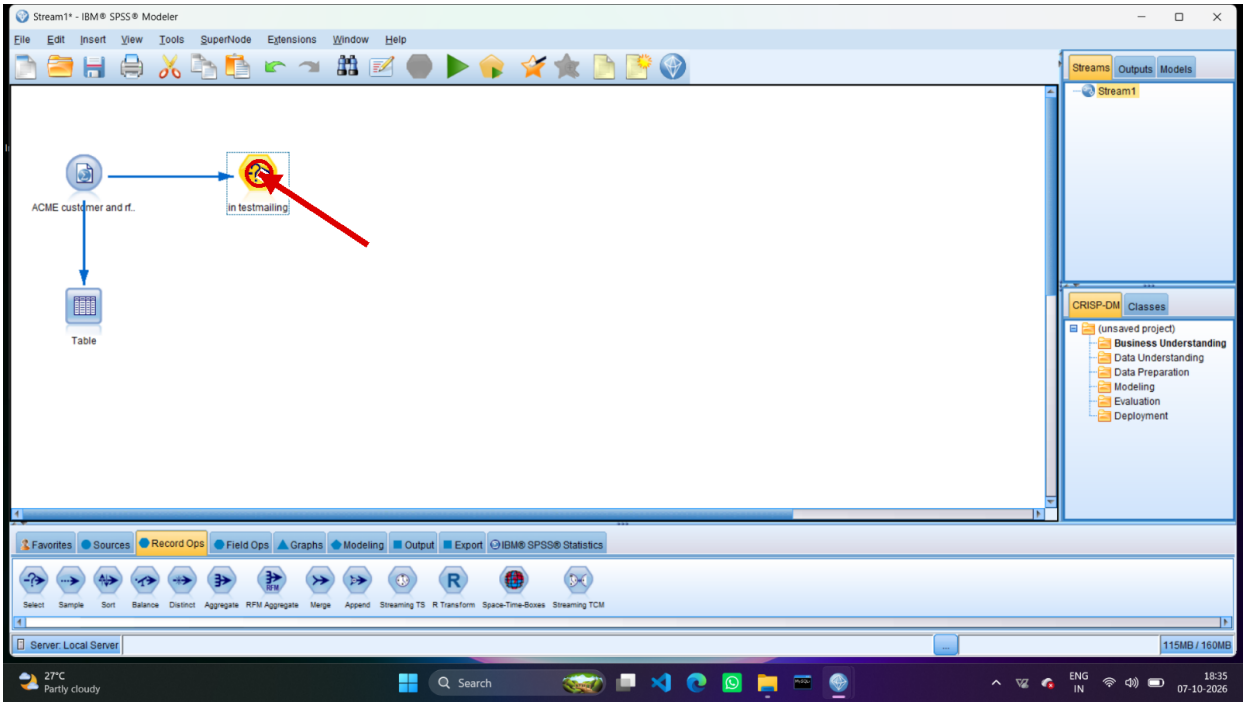
Step 10. Add a Select node

Use the Select node from Record Ops and connect it to the imported source. This node is used to keep only the customers belonging to the test mailing.



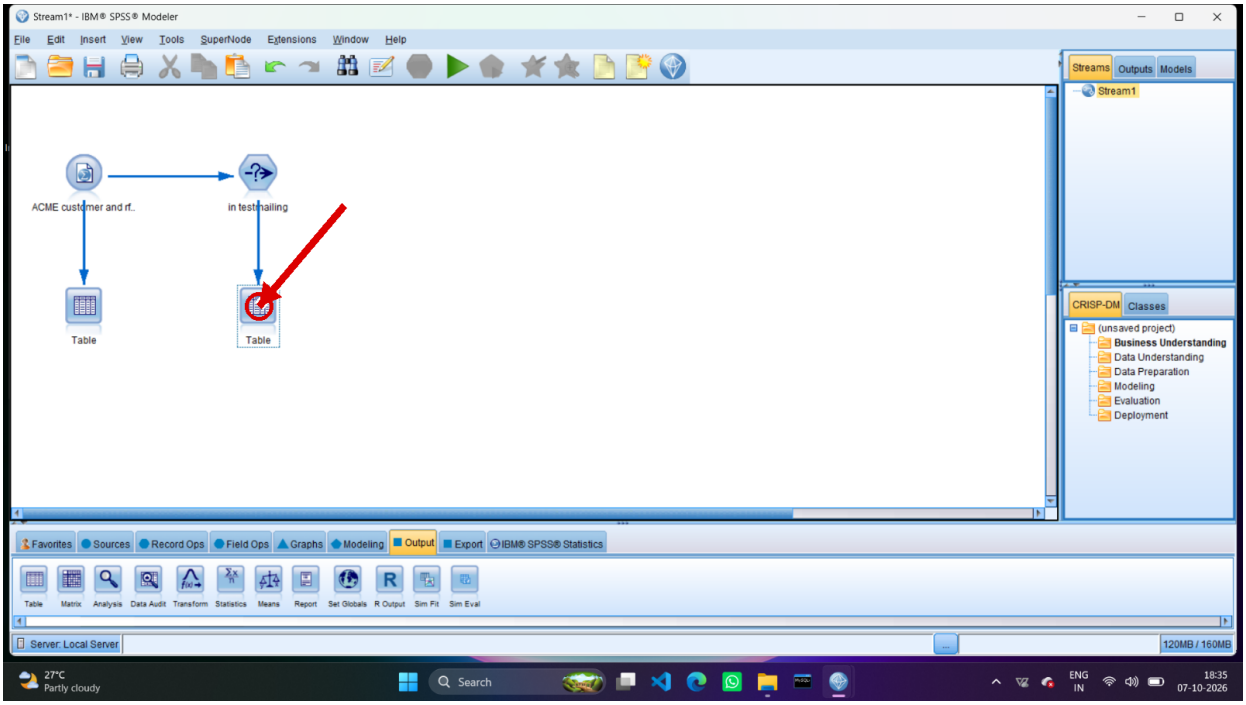
Step 11. Identify the test-mailing selection

Configure the Select node so the test-mailing records are isolated, then rename it clearly. The screenshot shows the node named for the test-mailing selection.



Step 12. Verify the selected records

Attach a Table node downstream from the Select node and run it. This provides a direct check that the intended subset is being passed forward.

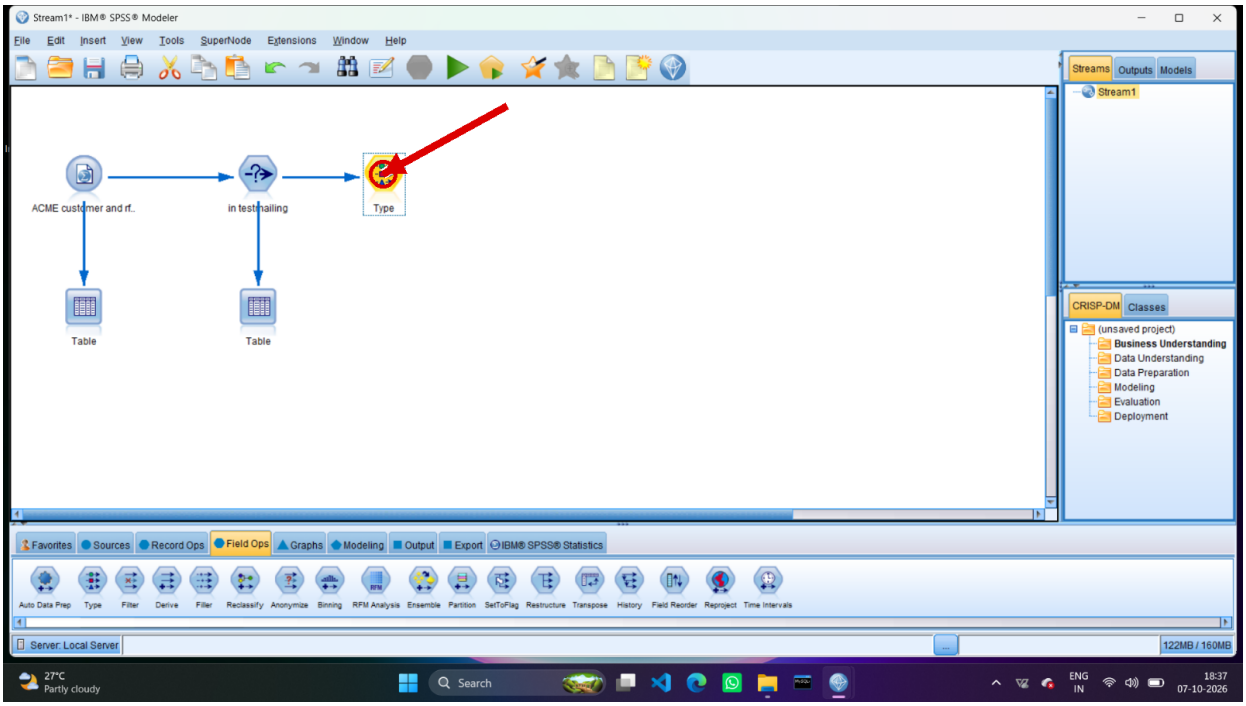


Task 4 — Build the CHAID model

The selected records are prepared with the correct field roles and then supplied to CHAID so the response target can be predicted from the chosen customer attributes.

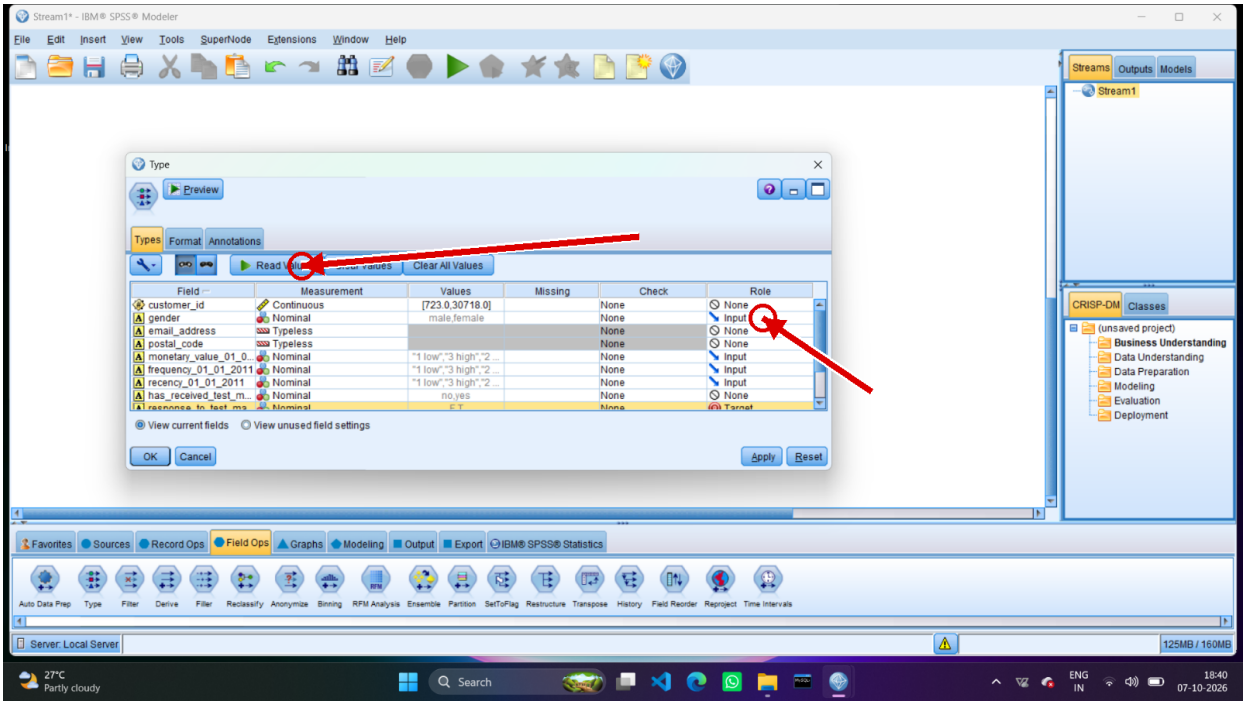
Step 13. Add the Type node

Place a Type node after the selected test-mailing data. The Type node is used to define field roles before the modelling algorithm is executed.



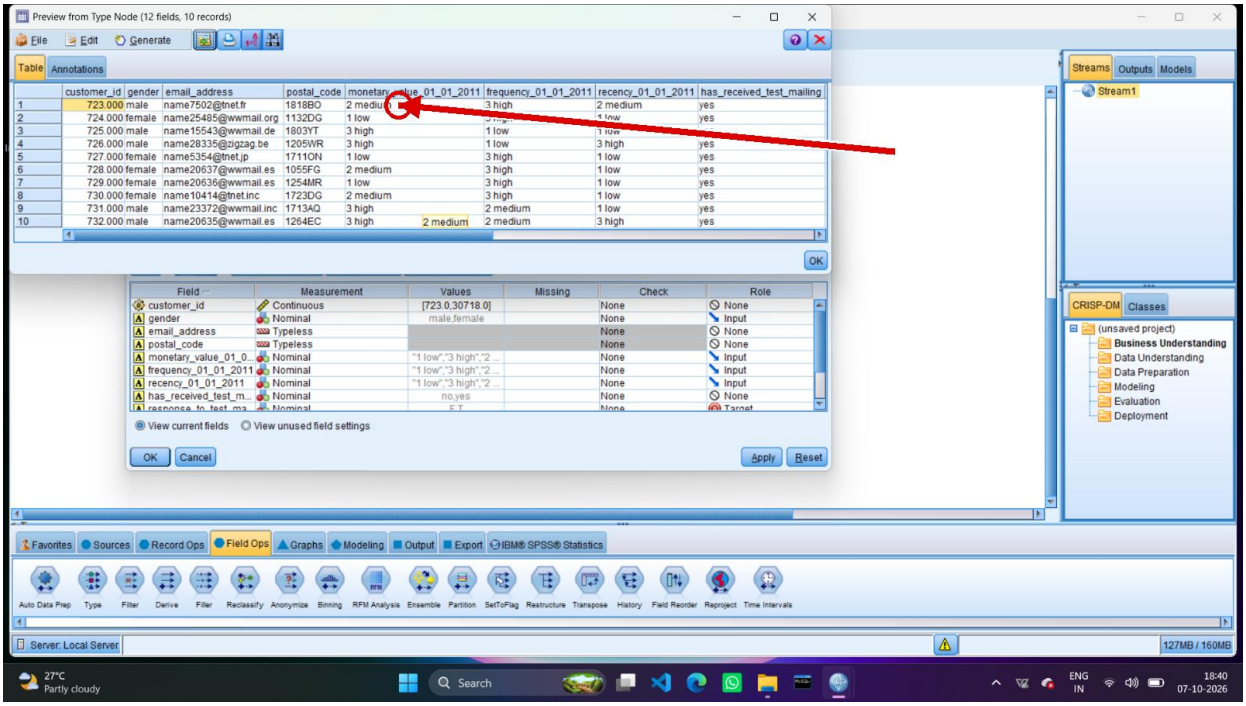
Step 14. Set field roles and read values

Use Read Values to load the field information, then keep gender and the three RFM fields as Input variables and response as the Target. The highlighted Role area confirms the assignment.



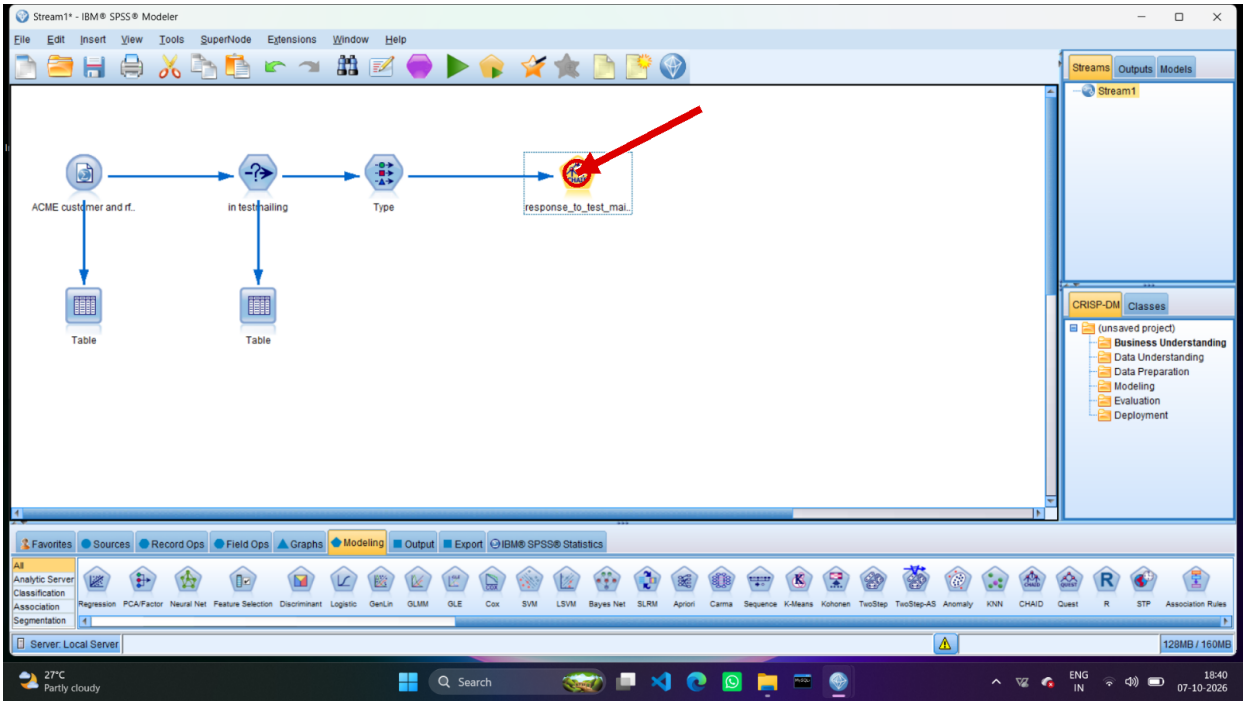
Step 15. Check the field values

Preview the Type node to confirm that the imported fields contain the expected categories and values. This is a quick validation before model execution.



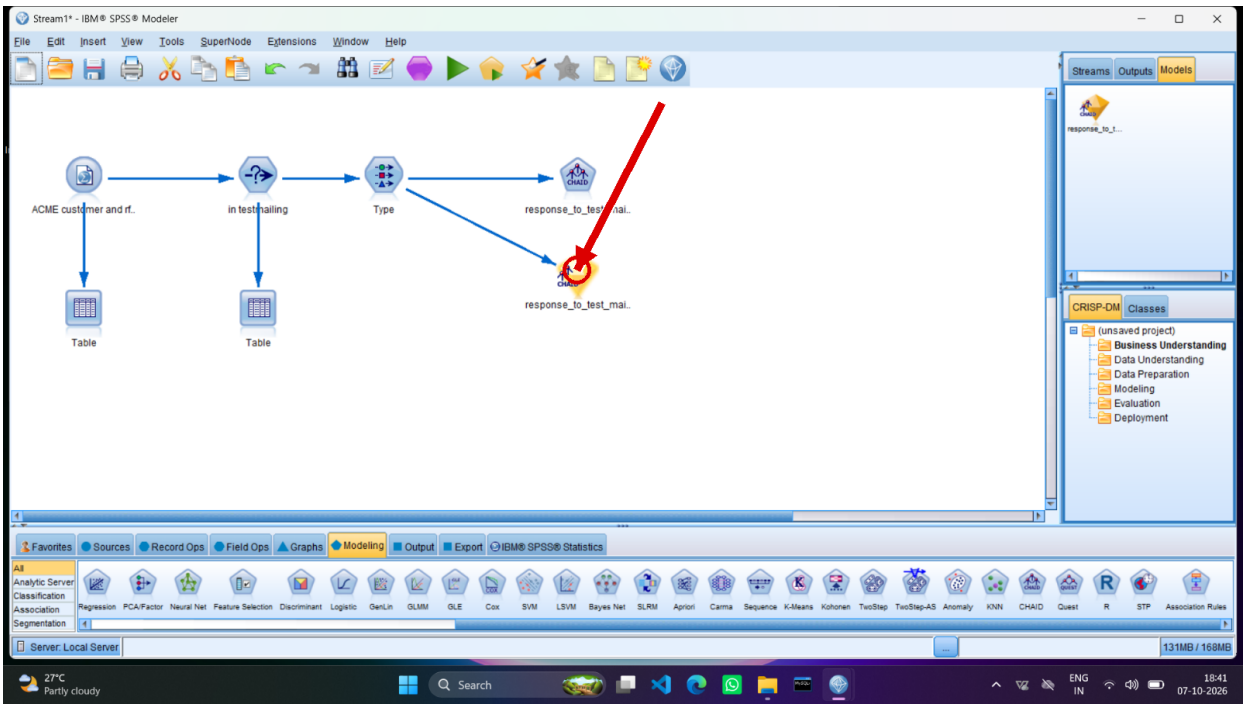
Step 16. Add the CHAID node

From the Modeling palette, add CHAID downstream from the Type node and connect it. The target field determines the model name shown in the stream.



Step 17. Run the CHAID model

Execute the CHAID node to generate the model nugget. The completed stream now contains the trained CHAID model that can be examined and reused for scoring.

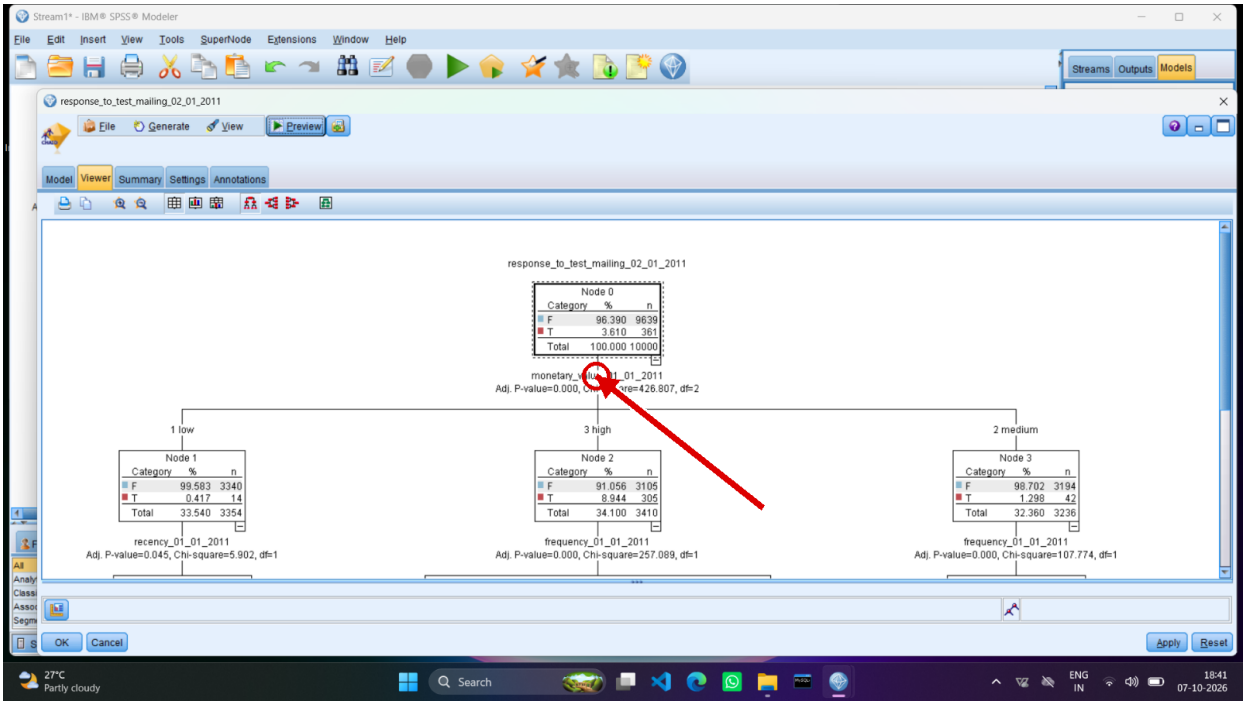


Task 5 — Examine the model

The generated tree is inspected to understand the first split and the customer groups associated with stronger response behaviour.

Step 18. Open and inspect the model viewer

Open the generated model nugget and view the tree structure. The first split shown in the screenshot is monetary_value_01_01_2011, which is the key split to inspect before moving to scoring.

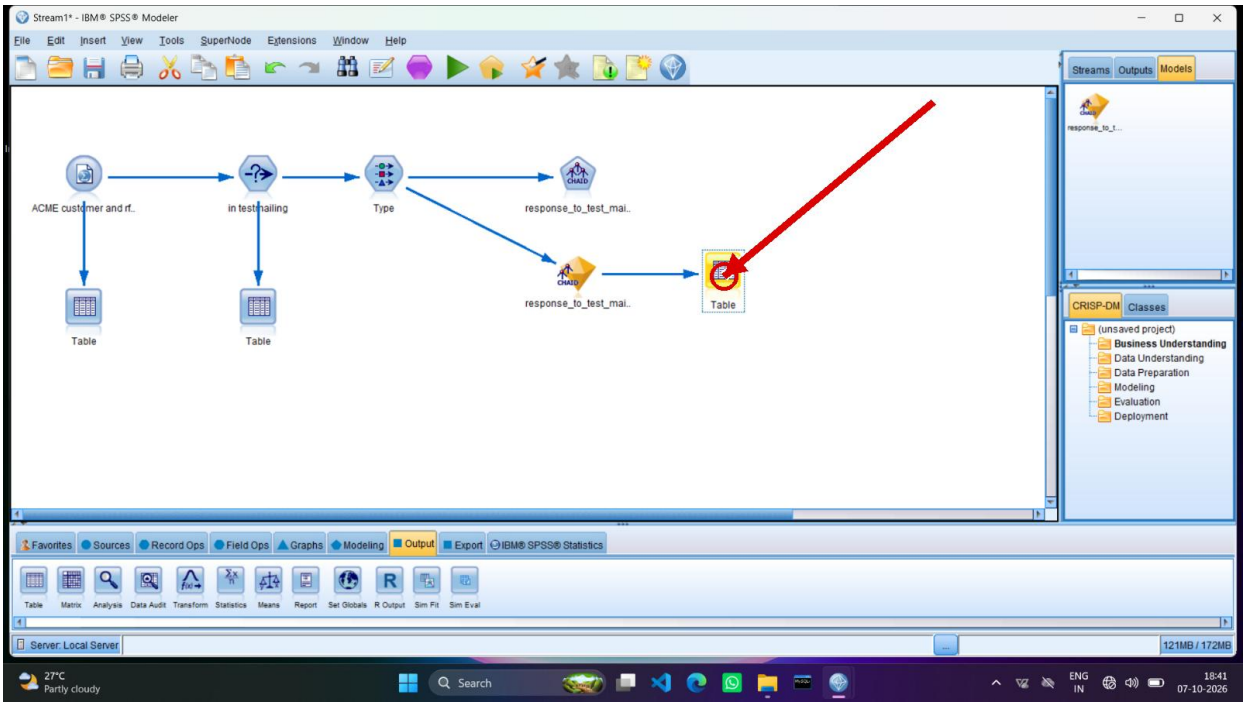


Task 6 — Interpret the fields added by the model

The model nugget adds prediction-related fields to the data. A Table node is used to see and interpret those generated fields.

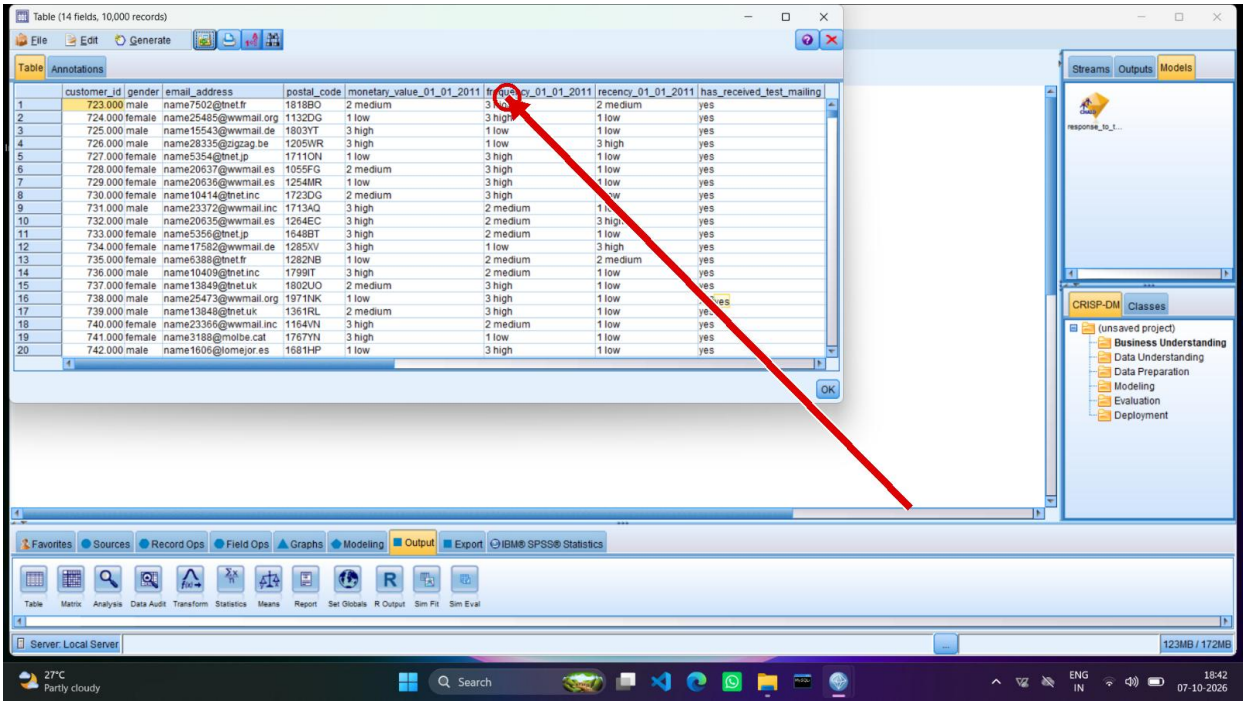
Step 19. Add a Table after the model nugget

Connect a Table node to the model nugget output to inspect the scored records. This makes the model-generated prediction fields available for interpretation.



Step 20. Check predicted response and confidence

Run the Table and inspect the added prediction and confidence fields. These values indicate the predicted class and the model's confidence for that prediction.

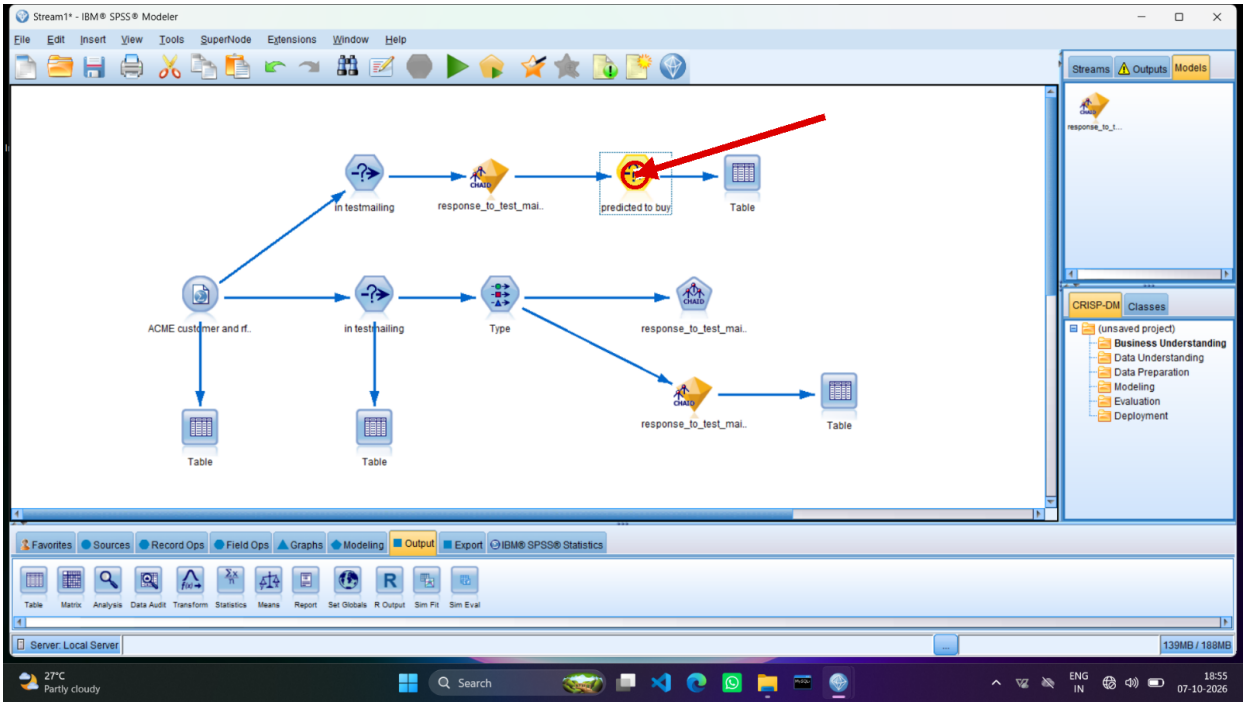


Task 7 — Use the model to score the remaining customers

The trained model is reused on customers outside the test mailing. The scoring branch identifies likely responders and prepares the required fields for export.

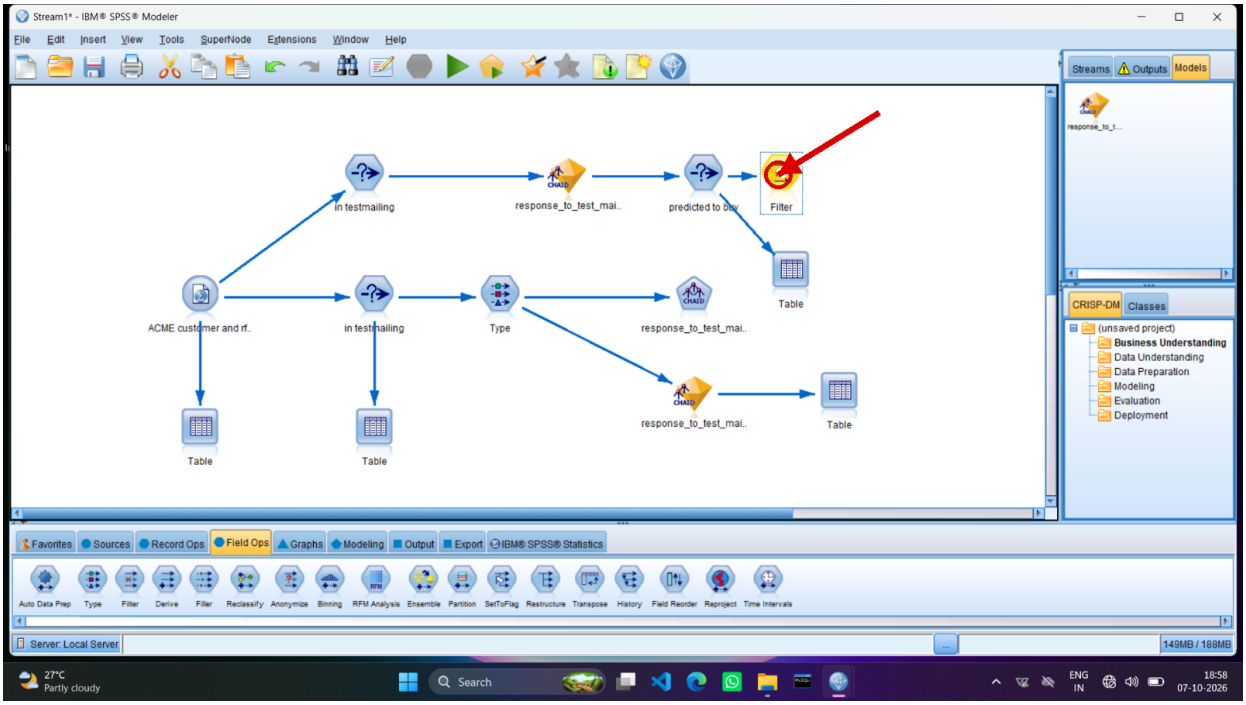
Step 21. Create the scoring branch

Copy the trained model nugget to the branch containing the customers outside the test mailing. Add a Select node to keep customers predicted to buy, then pass the result to a Table for checking.



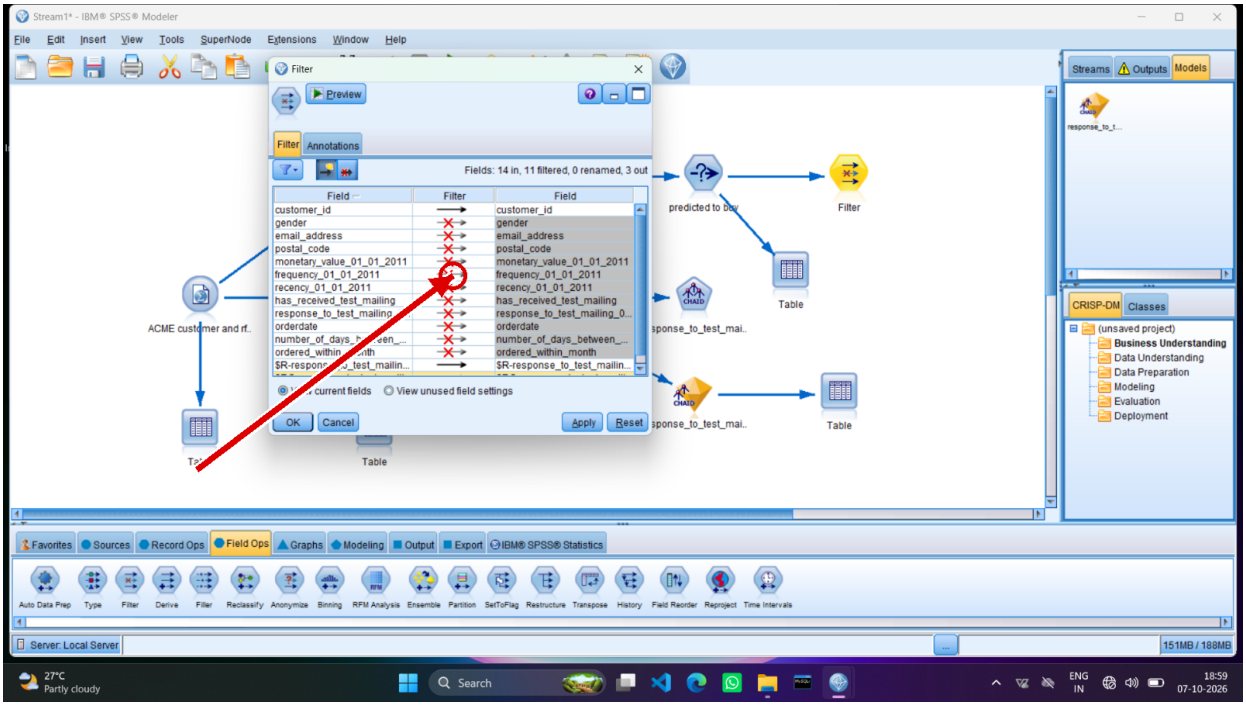
Step 22. Add the Filter preparation

Continue from the predicted-to-buy subset and add a Filter node. This prepares the scored data by keeping the fields needed for the final export.



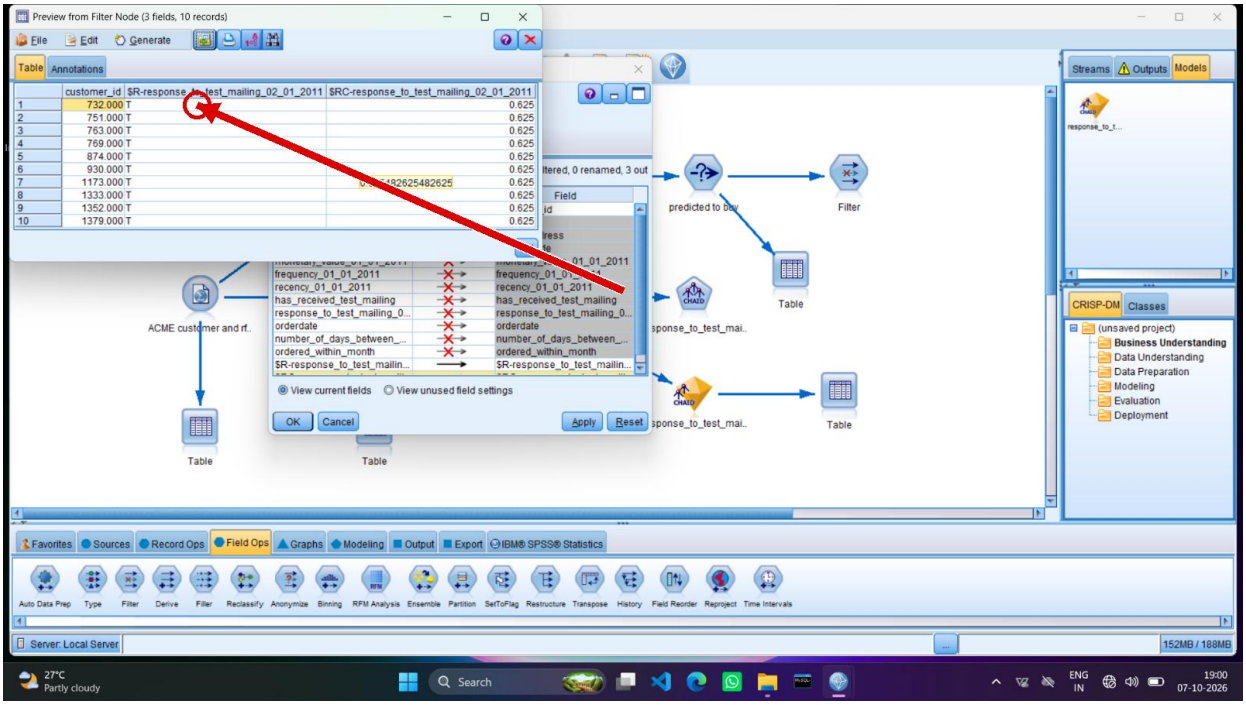
Step 23. Configure the Filter node

In the Filter settings, remove the fields that are not required and retain customer_id together with the model prediction and confidence fields. This keeps the final dataset compact.



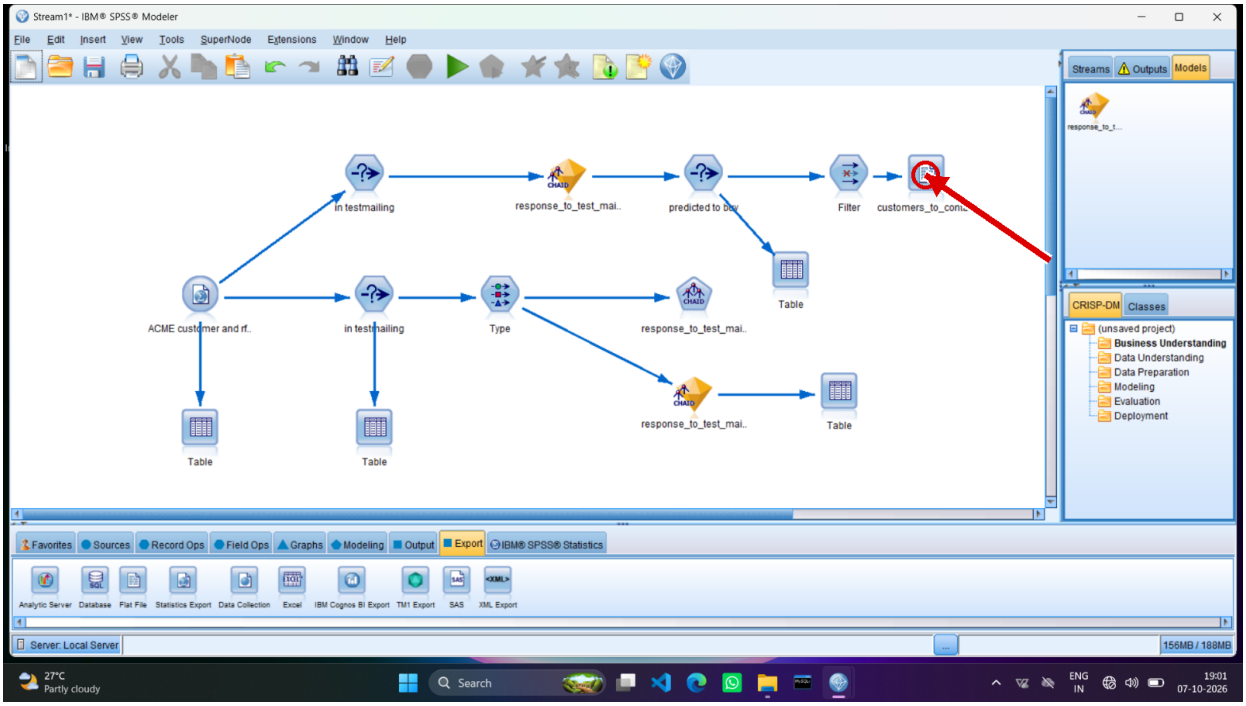
Step 24. Verify the filtered result

Run the downstream Table and confirm that the remaining records contain the customer identifier plus the prediction and confidence information. The screenshot shows the three-field output.



Step 25. Add the final export node

Use the Export palette to add a Flat File node after the Filter. Set the output name to customers_to_contact and connect it to the filtered stream.

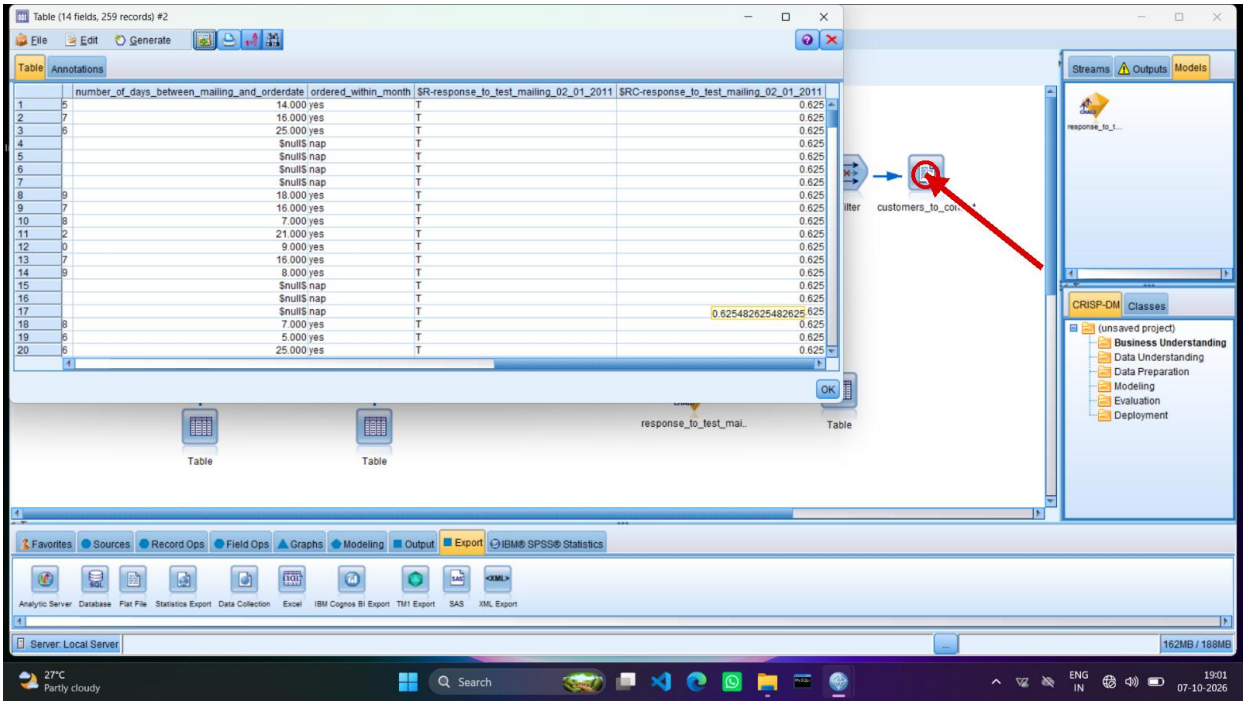


Task 8 — Export the results

The final filtered customer list is connected to a Flat File export so the selected contacts can be saved in the required output format.

Step 26. Verify the exported contact data

Run the final output path and inspect the resulting table/output. The screenshot shows the customers_to_contact result and the prediction/confidence columns used for the final contact list.



Task 9 — Exit IBM SPSS Modeler

Once the result has been checked, the application can be closed without saving any additional changes.